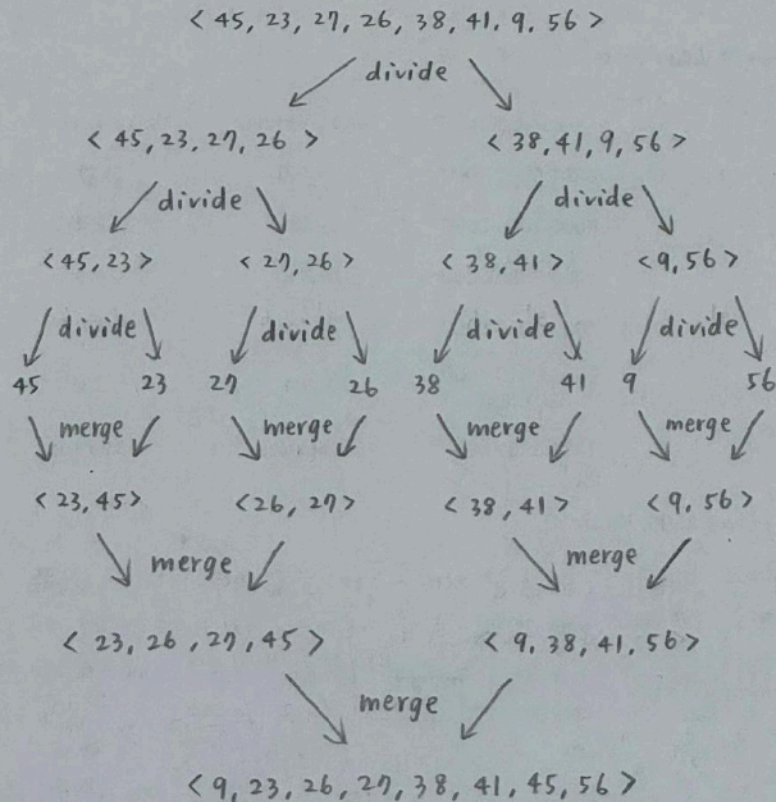


1. Consider the Merge-Sort:

- (a) Illustrate the operation of MERGE-SORT on the array $A = \langle 45, 23, 27, 26, 38, 41, 9, 56 \rangle$.
- (b) What is the best-case running time of MERGE-SORT?
- (c) What is the worst-case running time of MERGE-SORT?

1.

(a)



(b)(c)

無論 worst-case 或是 best-case 皆需進行 divide、conquer 和 combine
 設 n 為 array 的大小, h 為遞迴層數

	共 h 層				
第 i 層	0	1	2	...	$h-1$
array size	n	$\frac{n}{2}$	$\frac{n}{4}$	...	$\frac{n}{2^{h-1}}$

$$\frac{n}{2^{h-1}} = 1 \Rightarrow n = 2^{h-1} \Rightarrow h = \lg n + 1$$

$$\text{total cost} = cn(\lg n + 1)$$

$$= cn \lg n + cn$$

$$= \Theta(n \lg n)$$

$$\text{worst-case} = \text{best-case} = \Theta(n \lg n) \quad \#$$

2. Consider the searching problem:

Input: A sequence of n numbers $A = \langle a_1, a_2, \dots, a_n \rangle$ and a value v .

Output: An index i such that $v = A[i]$ or the special value NIL if v does not appear in A .

Write pseudocode for *linear search*, which scans through the sequence, looking for v . Using a loop invariant, prove that your algorithm is correct. Make sure that your loop invariant fulfills the three necessary properties.

2. Linear-search(A, v)

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1. for  $i \leftarrow 1$  to  $n$ 
2.   if  $A[i] == v$ 
3.     return  $i$ 
4. return NIL
```

證明:

Loop invariant: 執行第 i 個 loop 前, $A[1] \sim A[i-1]$ 都不等於 v

Initialization: $i=1$, loop 開始前, A 尚未被檢查

Maintenance: 設 $i=k$,
loop 開始前, $A[1] \sim A[k-1]$ 皆不等於 v
loop 開始後, 若 $A[k]$ 等於 v , 則程式立刻結束並回傳 index k , 若不相等, 則進入下一個 loop, 且 $A[1] \sim A[k]$ 都不等於 $v \Rightarrow$ loop invariant 被維持。

Termination: 程式結束時, i 分 2 種情況。

情形 1: $i \neq n+1$, 則 $v = A[i]$, 找到 v

情形 2: $i = n+1$, 根據 loop invariant, $A[1] \sim A[n]$ 都不等於 v , 因此 v 不存在。

程式的執行, 在以上 2 種情況下皆正確。